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Classifying Endurance Runners by Kinematics Strengthens Relationships of Muscle-Tendon Unit Function with Running Economy and Performance

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ABSTRACT

EIHARA Y., M. KINOSHITA, K. JOTOKU, S. MAEO, H. KANEHISA, and T. ISAKA. Classifying Endurance Runners by Kinematics Strengthens Relationships of Muscle-Tendon Unit Function with Running Economy and Performance. *Med. Sci. Sports Exerc.,* Vol. 58, No. 9, pp. 2087–2100, 2026. **Purpose:** Running kinematics and sex may alter the primary muscle and tendon demands for propulsion and shock absorption, potentially modifying how the muscle-tendon unit (MTU) function relates to the energy cost of running (ECR) and performance. This study examined whether clustering runners by running kinematics strengthens these relationships and how they differ by sex. **Methods:** One hundred twenty endurance runners (60 females) underwent assessments of MTU function, including slow (30° s^{-1}) and fast (180° s^{-1}) isokinetic knee extension and flexion torque and performance in countermovement, drop, and rebound jumps. These variables were summarized using principal component analysis. Running kinematics and ECR were assessed at 11 $km \cdot h^{-1}$ and 80% of $v\dot{V}O_{2max}$, and the season's best time (World Athletics score) was recorded. Runners were clustered via k-means based on foot-strike angle and knee extension/flexion range of motion. Stepwise regression analyses tested whether MTU-related associations differed between cluster-specific and whole-cohort models. **Results:** The k-means identified three or four clusters per sex at each running velocity condition. In whole-cohort analyses for each sex, only weak to fair correlations were found between MTU function and either energy cost ($r = -0.204$ to 0.267) or World Athletics score ($r = -0.276$ to 0.175). Clustering significantly ($P < 0.05$) strengthened these relationships in a cluster-specific manner (adjusted $R^2 = 0.159$–0.595). This pattern was more evident at 80% of $v\dot{V}O_{2max}$, particularly in male clusters. Across conditions, MTU-related variables were selected in a greater proportion of female clusters (57.1%) than male clusters (31.2%). **Conclusions:** Kinematic-based clustering strengthens sex- and cluster-specific associations of MTU function with the ECR and performance, supporting more individualized performance enhancement strategies. **Key Words:** CLUSTER ANALYSIS, ENERGY COST OF RUNNING, JUMP PERFORMANCE, LONG-DISTANCE RUNNING, MUSCLE STRENGTH

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Participation in running events has expanded markedly, from 1.8 million in 2001 to 8 million in 2018 (1). Success in endurance running is largely determined by aerobic parameters such as running economy and maximal oxygen uptake ($\dot{V}O_{2max}$) (2). However, increasing $\dot{V}O_{2max}$ in trained endurance runners can be challenging (3). In contrast, running economy, defined here as the metabolic or energy cost of running (ECR) at a given speed (4), is influenced by modifiable factors including muscle-tendon unit (MTU) function (5,6), running biomechanics (7–9), and physiological characteristics (10–12). Among these, MTU function, such as muscle strength and jump performance, can be improved relatively easily by incorporating strength training into running programs (13). Accordingly, many studies have investigated factors affecting running economy and performance, as well as strength training modalities that optimize these outcomes (14,15).

Lower-limb muscles, in interaction with tendon behavior, are essential for achieving superior running economy and performance. The quadriceps femoris attenuates impact forces at touchdown (16,17), while the hamstrings contribute to propulsion during the late stance (18,19). Co-contraction of these muscle groups increases knee and leg stiffness (20,21), enabling the efficient use of impact forces through the stretch-shortening cycle (SSC) (5) and, consequently, improvement in running economy (22). Furthermore, interaction between the triceps surae and Achilles tendon during running mirrors that seen in jumping (23), and such interaction has been linked to economical running (12,24). Indeed, resistance and plyometric training aimed at enhancing muscle strength and jump performance can improve running economy and performance (13,25,26).

Nevertheless, responses to resistance and plyometric training aimed at improving running economy vary considerably among individuals (27), suggesting that the importance of specific MTU function for running economy and performance differs between runners. This interpretation is supported by previous evidence indicating that relationships between movement patterns and performance outcomes may differ across subgroups (22). Sex is one factor likely to influence these relationships between MTU function and both running economy and performance. Female runners often (28,29), but not always (30), exhibit lower knee flexor and higher knee extensor activation than males when running at the same absolute velocity (e.g., 12 $km \cdot h^{-1}$). This pattern is thought to reflect greater knee flexion (KF) angle and increased quadriceps activity at touchdown in females (31–33). Consequently, the association between knee extension (KE) torque and running economy or performance may be more pronounced in females than in males.

Even within the same sex, running kinematics may influence which MTU functions are most relevant to running economy and performance. Rearfoot strikers show greater biceps femoris activation than forefoot strikers at the same absolute velocity (34–36), and runners with a larger knee extension/flexion range of motion (KE/KF ROM) activate the rectus femoris more during propulsion (27). Conversely, forefoot strikers with smaller KE/KF ROM tend to rely more on SSC utilization (i.e., effective elastic energy storage and return and stretch-related potentiation) of the lower leg MTU (20,35). Therefore, grouping runners with similar kinematic profiles (e.g., foot-strike angle and KE/KF ROM) may strengthen the observed relationships of MTU function with running economy and performance. For example, clusters characterized by a dorsiflexed foot strike (rearfoot) may show stronger associations between KF strength and running economy and/or performance.

The k-means clustering method groups participants based on multivariate profiles into distinct clusters and is particularly useful when the same set of variables is measured for all participants (e.g., running kinematics), enabling pattern-based (parallel) clustering across individuals (37,38). In this study, we applied k-means clustering to classify endurance runners according to foot-strike angle and KE/KF ROM at the same running velocity. Because racing speeds vary between individual runners, we examined relationships between MTU function and running economy and/or performance at both the same absolute speed (11 $km \cdot h^{-1}$) and the same relative speed (80% of the velocity of $\dot{V}O_{2max}$ [$v\dot{V}O_{2max}$]). The purposes of this study were 1) to examine relationships of MTU function with running economy and performance in the whole cohort and within each kinematic cluster, and 2) to determine how these relationships differ between sexes at absolute and relative velocities. We hypothesized that 1) clustering would strengthen these relationships compared with nonclustered analyses, and 2) clusters showing strong KE strength-economy and/or performance relationships would be more prevalent in females than in males.

METHODS

Participants

One hundred twenty trained (Tier 2–3) (39) endurance runners (60 females) voluntarily participated in this study (Table 1). Participants were required to meet the following inclusion criteria: 1) age 18–50 years; 2) season's best time in a 5 km or longer race with World Athletics (WA) points of > 54 for males (e.g., full-marathon < 4:00:05) and > 259 for females (e.g., full-marathon < 4:30:02) (40); 3) body mass index < 24 $kg \cdot m^{-2}$; 4) no musculoskeletal injuries requiring modified training within the past 3 months and no injuries within the past month; and 5) running more than twice per week during the preceding 3 months. Running performance was expressed as the WA score converted from the season's best race time. Participants were stratified by age group (18–30, 31–40, and 41–50 years), with 40 participants per group and a

TABLE 1. Participant characteristics.

	Male (n = 60)	Female (n = 60)
Age (years)	34.1 ± 10.3	34.1 ± 9.7
Height (m)	1.71 ± 0.06	1.59 ± 0.05
Body mass (kg)	60.0 ± 5.9	49.4 ± 5.9
Fat percentage (%)	11.2 ± 3.4	19.5 ± 5.3
WA score	527.7 ± 224.7	618.4 ± 183.2
Energy cost at 11 $km \cdot h^{-1}$ ($kcal \cdot kg^{-0.915} \cdot km^{-1}$)	1.48 ± 0.13	1.47 ± 0.13
Energy cost at 80% of $v\dot{V}O_{2max}$ ($kcal \cdot kg^{-0.839} \cdot km^{-1}$)	1.96 ± 0.13	1.98 ± 0.14
$\dot{V}O_{2max}$ ($mL \cdot kg^{-1} \cdot min^{-1}$)	55.4 ± 5.3	46.3 ± 5.5
$v\dot{V}O_{2max}$ ($km \cdot h^{-1}$)	18.1 ± 1.6	15.5 ± 1.1

Data are presented as mean ± SD.
The seasonal record over distances from 5000 m to the ultra-marathon (100 km) was converted to a WA score.

1:1 male-to-female ratio. This study was approved by the ethics committee of Ritsumeikan University (BKC-LSMH-2023-011) and conducted in accordance with the Declaration of Helsinki. All participants provided written informed consent after being informed of the study purpose, procedures, and potential risks.

Experimental Design and Procedure

Participants attended two laboratory sessions separated by 24–168 h, under controlled ambient temperature (21.0°C–22.7°C). They refrained from strenuous exercise and alcohol for 24 h, and caffeine for 6 h before each visit. The following variables were measured: Session 1) body composition, KE/KF strength, jump performance, and $\dot{V}O_{2max}$; Session 2) body composition, running kinematics, and ECR. Season's best performance within the past year was obtained from official records. Body composition data from Session 1 were used to scale $\dot{V}O_{2max}$, and from Session 2 to normalize ECR.

Body composition. Height and body mass were measured using an automatic scale with a stadiometer (WB-510, TANITA Corp., Tokyo, Japan). Body fat percentage, muscle mass, and fat-free mass were measured using a multifrequency, segmental bioelectrical impedance analyzer (InBody770, InBody Japan Co., Ltd., Tokyo, Japan), which has demonstrated good validity compared with dual-energy X-ray absorptiometry (41,42).

KE/KF torque. Isokinetic KE/KF concentric torque of the right leg was assessed in a seated position (hip flexion 85°) using a dynamometer (Biodex System IV, Biodex Medical Systems, Shirley, NY). The seat was adjusted to align the lateral tibial condyle with the dynamometer axis of rotation, and the trunk and tested thigh were secured with straps. Gravity correction was applied for the static position at approximately 45° of KF. Participants performed a warm-up of three isometric KE/KF contractions at 90° KF at 50%, 70%, and 90% of maximal effort, followed by practice isokinetic contractions in both slow (30° s^{-1}) and fast (180° s^{-1}) conditions. These practice trials consisted of 2–3 complete cycles of KE and KF, from 90° KF to 0° full extension and back to 90° flexion, performed at 50% effort. Maximal isokinetic concentric torque was then measured during two complete cycles at each angular velocity, with a rest period of 60 s between velocity conditions. The highest torque value for KE and KF at each velocity was used for analysis.

Jump performance. Countermovement jump (CMJ), 30 cm drop jump, and three consecutive rebound jump tests were performed three times each. Jump height and ground contact time were recorded using an optical measurement system (OptoJump Next, Microgate, Bolzano, Italy), which has been validated previously (43,44). For the CMJ, participants performed a rapid squat followed by a maximal vertical jump with an arm swing. The highest jump height was used for analysis. For the drop jump, participants stood on a 30 cm box, stepped off, landed, and rebounded to maximal height with an arm swing while minimizing ground contact time. For the rebound jumps, participants performed three consecutive maximal vertical jumps with an arm swing and minimal contact time between jumps. For both the drop and rebound jumps, a jump index (jump height divided by contact time) was calculated, and the trial with the highest jump index was used for analysis. If two or three trials showed the same highest jump index, the trial with the greatest jump height was selected.

$\dot{V}O_{2max}$. $\dot{V}O_{2max}$ was measured on a treadmill with no incline (Valiant Ultra; Lode BV, Groningen, Netherlands) using a breath-by-breath gas analyzer (AE-310s, Minato Medical Science, Osaka, Japan) and masks (AMA104 and AMA512, Minato Medical Science, Osaka, Japan). Heart rate was recorded using a heart rate sensor (Polar H10, Polar, Kempele, Finland). The gas analyzer was calibrated before each session with reference gases (16% O_2 and 5% CO_2). The repeatability of measurements with this breath-by-breath gas analysis system has been confirmed and is reported in Supplemental Table 1, Supplemental Digital Content, https://links.lww.com/MSS/D402. As a warm-up, participants ran on the treadmill at 8 and 11 $km \cdot h^{-1}$ for 4 min each, separated by a 2-min rest, consisting of a 1-min walk at 3.2 $km \cdot h^{-1}$ followed by 1 min of passive rest. After an additional 2-min rest, participants performed a continuous incremental test starting at 8 $km \cdot h^{-1}$, with velocity increased by 1 $km \cdot h^{-1}$ every 1-min up to 14 $km \cdot h^{-1}$, and thereafter by 0.6 $km \cdot h^{-1}$ until volitional exhaustion, defined as the inability to maintain the required running velocity. $\dot{V}O_{2max}$ was defined as the highest 30-s moving average. Attainment of $\dot{V}O_{2max}$ was confirmed if all of the following criteria were met: heart rate ≥ 90% of age-predicted maximum (220 −age); respiratory exchange ratio ≥ 1.10 (26); and an increase of $\dot{V}O_2$ ≤ 1 $mL \cdot kg \cdot min^{-1}$ with increased workload (45). All participants completed the test until volitional exhaustion, and all predefined criteria were met. The velocity associated with $\dot{V}O_{2max}$ ($v\dot{V}O_{2max}$) was calculated by accounting for fractional completion of the final stage (46). For example, if a participant completed 30 s of the final stage at 13 $km \cdot h^{-1}$, $v\dot{V}O_{2max}$ was calculated as 13.5 $km \cdot h^{-1}$ (13 [velocity of the final stage] + 1 [increment rate] × 30/60 [fractional time]).

Energy cost and allometric scaling. The ECR was measured on a 0% gradient treadmill (Aspire; Life Fitness, Rosemont, IL) using the same gas analyzer and mask as for the $\dot{V}O_{2max}$ measurement, at 11 $km \cdot h^{-1}$ and 80% of $v\dot{V}O_{2max}$. A 0% incline was selected because recent evidence indicates that, at submaximal running speeds, both the ECR (47) and running biomechanics (48) are comparable between 0% and 1% treadmill inclines. Participants

ran for 4 min at each velocity. The 11 km·h^{-1} velocity was chosen because all runners in a previous study (n = 97), including females with a wide performance range, were able to run at this velocity with respiratory exchange ratio ≤1.00 (7). The 80% of $v\dot{V}O_{2max}$ condition was selected because it can theoretically be sustained for > 2 h without rest (49) and predicts 16 km running time (2). ECR (kcal·kg^{-1}·km^{-1}) was calculated as the average during the final minute of each trial/velocity using second-by-second $\dot{V}O_2$ and $\dot{V}CO_2$ data. Steady-state $\dot{V}O_2$ at both velocities was confirmed by a low coefficient of variation (4.82% ± 1.56% at 11 km·h^{-1} and 4.12% ± 1.60% at 80% of $v\dot{V}O_{2max}$) and the absence of systematic drift in the relationship between time and $\dot{V}O_2$ (slope: 0.002 ± 0.043 at 11 km·h^{-1}; −0.0002 ± 0.047 at 80% of $v\dot{V}O_{2max}$). Substrate utilization (g·min^{-1}) was estimated using updated non-protein respiratory quotient equations (50). The energy derived from each substrate was then calculated by multiplying fat and carbohydrate usage by 9.75 and 4.07 kcal, respectively, reflecting their average energy content during moderate- to high-intensity exercise (51). The total energy expenditure was expressed in kcal·km^{-1}. ECR values were allometrically adjusted for body mass. The allometric exponent was determined by transforming the equation $B = aM$b (52) into its logarithmic form: log (B) = log (a) + b·log (M), where B is metabolic rate, a is a constant, M is body mass, and b is the exponent. Parallelism of regression lines between ECR and body mass was tested using an analysis of covariance (ANCOVA) by sex and showed no sex effect (P = 0.608–0.755). Power functions provided a better fit than linear models based on R^2 and root mean square error. Accordingly, the allometric exponent was obtained for all participants as b = 0.915 (kcal·kg$^{-0.915}$·km^{-1}) for 11 km·h^{-1} and b = 0.839 (kcal·kg$^{-0.839}$·km^{-1}) for 80% of $v\dot{V}O_{2max}$.

Running kinematics. Thirty-two reflective markers were placed on anatomical landmarks. Three-dimensional kinematic data during running at 11 km·h^{-1} and 80% of $v\dot{V}O_{2max}$ were collected using a 21-camera motion capture system operating at 250 Hz (MAC3D, Motion Analysis Corporation, Rohnert Park, CA) and Cortex software (Motion Analysis Corporation, Santa Rosa, CA). Marker trajectories were filtered with a bidirectional fourth-order low-pass Butterworth filter (cut-off frequency: 15 Hz) to remove high-frequency noise (53). Ten consecutive strides, recorded 3 min after the onset of running, were analyzed. Touchdown and toe-off were identified using a two-step procedure. First, analysis windows were defined using kinematic criteria. Touchdown was initially identified as the instant when the anterior-posterior velocity of the heel marker changed from positive to negative, ending at the subsequent minimum in vertical heel position. Toe-off was initially identified as the point when the vertical toe position (midpoint between the first and fifth metatarsal heads) exceeded 0.1 m or reached its maximum height (7,54–56). Second, the exact instants of touch-down and toe-off were determined within these windows using acceleration-based measures. The onset of touchdown was identified by the first peak in vertical acceleration of either the heel or first metatarsal head markers, accommodating different foot-strike patterns and footwear (55,56). Toe-off was identified as the time point of peak vertical jerk (first derivative of acceleration) of the toe marker.

Kinematic variables were averaged over 10 steps for the right leg at each running velocity. If a marker became detached during running, its position was estimated using linear interpolation. Foot-strike angle in the sagittal plane was determined from the initial joint angle at touchdown (the sagittal foot-ground angle), and treated as a continuous variable, classified as relatively dorsiflexed or plantarflexed, without categorizing runners as rearfoot or forefoot strikers, in accordance with a previous study (57). KE/KF (the sagittal thigh-shank angle) ROM was calculated as the difference between the maximum and minimum knee joint angles during the stride. In addition, mean KE/KF angular velocities during the stance phase at both 11 km·h^{-1} and 80% of $v\dot{V}O_{2max}$ were calculated.

Statistical Analysis

All statistical analyses were conducted using RStudio (version 2025.09.2+418, Boston, MA).

Principal component analysis. Muscle strength and jump performance variables were summarized using principal component analysis (PCA). As a result, four principal components (PCs) were extracted for each sex, which are described in detail in the Results section.

K-means clustering. Foot-strike angle in the sagittal plane and KE/KF ROM were standardized into Z-scores for each sex. The ECR, the foot-strike angle, and KE/KF ROM at 80% of $v\dot{V}O_{2max}$ were adjusted by individual running velocity using ANCOVA. The k-means clustering method was applied separately for each sex and each running velocity, using these two kinematic variables (foot-strike angle and KE/KF ROM). Although multicollinearity is generally less critical in k-means clustering, we nonetheless verified that the kinematic variables used for clustering were not strongly correlated (r = −0.164 to 0.115). Accordingly, no additional control for multicollinearity was required. The optimal number of clusters was determined using the Elbow method: a linear reference line was drawn between cluster numbers 1 and 10, and the perpendicular distance from this line was calculated (Fig. 1) (58). The number of clusters corresponding to the greatest perpendicular distance was selected. After clustering, outliers were removed using the Distance-Based Algorithm (59) (male: n = 6, female: n = 3 at 11 km·h^{-1}; male: n = 3, female: n = 2 at 80% of $v\dot{V}O_{2max}$).

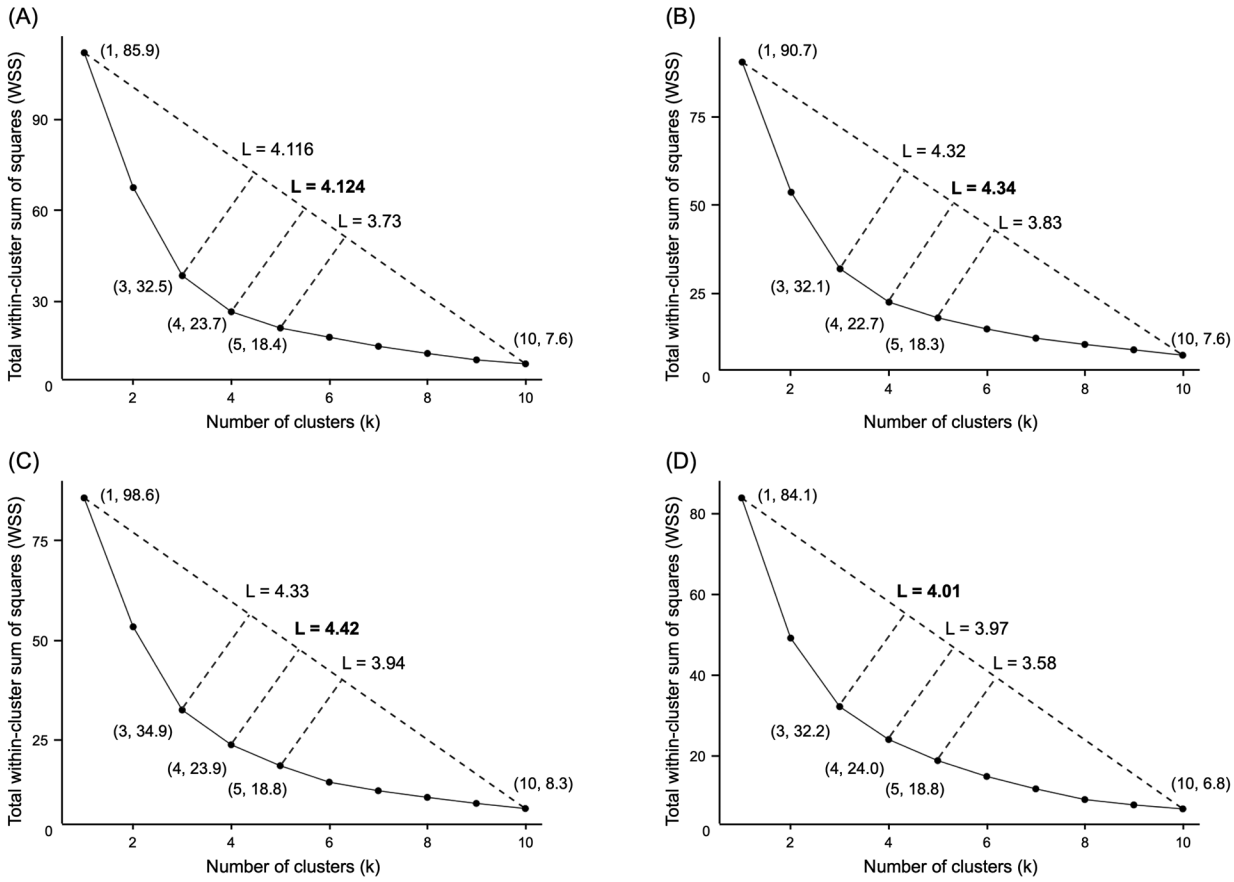


FIGURE 1—Elbow analysis for determining the optimal number of clusters (A–D). Panels represent male runners at 11 km/h (A), female runners at 11 km/h (B), male runners at 80% of $v\dot{V}O_{2max}$ (C), and female runners at 80% of $v\dot{V}O_{2max}$ (D). The graph shows the within-cluster sum of squares (WSS) for cluster numbers 1, 3–5, and 10. "L" represents the length of the perpendicular line, indicating the optimal number of clusters based on the longest distance.

Correlation analyses. Normality was assessed using Shapiro–Wilk test. When a variable did not meet normality assumptions, Spearman rank correlation was used to calculate relationships between MTU functions (KE/KF torques and jump performances) and running performance-related variables (ECR and WA score) within each cluster. Correlation coefficients were interpreted as weak ($0.0 < |r| < 0.25$), fair ($0.25 \le |r| < 0.50$), moderate ($0.50 \le |r| < 0.75$), or strong ($0.75 \le |r| < 1.00$) (60). Relationships with $|r| \ge 0.5$ (moderate or strong) were considered meaningful (61). To test whether the cluster strengthened these relationships compared with the nonclustered (whole) cohort, correlation coefficients were compared using the Williams test (62).

Regression analyses. Stepwise regression analyses were performed, entering MTU function variables (PC1–4) as independent variables, and ECR or WA score as dependent variables in each cluster. In addition, the relative contribution of each retained PC was quantified using the Lindeman–Merenda–Gold method. To examine whether the relationships between MTU function and outcomes differed between clustered and nonclustered runners, linear regression slopes within each cluster were statistically compared with those obtained from the whole cohort (nonclustered runners) using interaction terms.

RESULTS

Principal Component Analysis

In male runners, PCA identified four components explaining 88.9% of the total variance in MTU function variables (PC1: 53.2%, PC2: 22.3%, PC3: 9.3%, PC4: 4.2%). PC1 loaded strongly on jump performance and moderately on muscle strength, representing overall MTU function (Strength & Jump [−]). PC2 was characterized by positive loadings on muscle strength and negative loadings on SSC ability (Strength [-]/SSC [+]). PC3 reflected a balance between knee extensor and flexor function, with negative loadings for KE and positive loadings for KF (KE [−]/KF [+]). PC4 loaded positively on CMJ height and negatively on jump indices, indicating nonreactive versus SSC-based jump performance (CMJ [+]/SSC ability [−]) (Table 2).

TABLE 2. Factor loadings of the PCA on MTU-related variables for males.

Variables	PC1	PC2	PC3	PC4
Isokinetic slow KE	–0.152	–0.509	–0.533	0.063
Isokinetic fast KE	–0.259	–0.422	–0.427	0.067
Isokinetic slow KF	–0.261	–0.426	0.461	–0.088
Isokinetic fast KF	–0.262	–0.402	0.524	–0.138
CMJ height	–0.368	0.170	0.140	0.801
DJ height	–0.407	0.151	–0.023	0.159
DJ index	–0.399	0.200	–0.062	–0.391
RJ height	–0.400	0.239	–0.115	–0.058
RJ index	–0.392	0.265	–0.095	–0.377
Standard deviation	2.188	1.416	0.914	0.613
Proportion of variance	0.532	0.223	0.093	0.042
Cumulative proportion	0.532	0.755	0.847	0.889
Interpretation	Strength & Jump (–)	Strength (–)/SSC ability (+)	KE (–)/KF (+)	CMJ (+)/SSC ability (–)

DJ, drop jump; KE, knee extension torque; KF, knee flexion torque; PC, principal component; RJ, rebound jump; SSC, stretch-shortening cycle.

In female runners, PCA identified four components explaining 89.2% of the total variance (PC1: 48.2%, PC2: 25.2%, PC3: 10.1%, PC4: 5.8%). PC1 loaded negatively on both muscle strength and jump performance, reflecting overall MTU function, with lower scores indicating greater MTU function capacity (Strength & Jump [−]). PC2 represented a continuum from strength-dominant to jump/SSC-dominant characteristics, with positive loadings on isokinetic knee torques and negative loadings on jump performance (Strength [+]/Jump [−]). PC3 reflected knee extensor versus flexor dominance (KE [+]/KF [−]). PC4 contrasted nonreactive jump performance with reactive SSC utilization, loading positively on CMJ height and KF torque and negatively on reactive jump indices and KE torque (CMJ & KF [+]/SSC & KE [−]) (Table 3).

Male Clusters at 11 km·h^{-1}

Participants were divided into four clusters in the 11 km·h^{-1} condition (Fig. 2). In the nonclustered (whole-cohort) analyses, only weak to fair relationships were observed between MTU function variables (KE/KF torque and jump performance) and running performance-related variables (ECR and WA score) (ECR, $r = -0.204$ to 0.162; WA, $r = -0.269$ to -0.001; see "ALL" columns in Supplemental Table 2, Supplemental Digital Content, https://links.lww.com/MSS/D402).

Cluster 1: Dorsiflexed landing. PC4 (higher SSC ability) was selected as a factor associated with lower ECR, although this model did not reach statistical significance (adjusted $R^2 = 0.142$, $P = 0.101$). In contrast, PC1 (higher overall MTU function), together with PC4 (higher SSC ability), significantly explained the variance in WA score (adjusted $R^2 = 0.284$, $P = 0.044$).

Cluster 2: Large KE/KF ROM. PC2 (higher muscle strength) and PC4 (higher SSC ability) were selected as explanatory variables for lower ECR, but the model did not reach statistical significance (adjusted $R^2 = 0.258$, $P = 0.124$).

Cluster 3: Plantarflexed landing. No variables were selected to explain the variance in ECR or WA score by the regression analysis.

Cluster 4: Small KE/KF ROM. No variables were selected to explain the variance in ECR or WA score by the regression analysis.

TABLE 3. Factor loadings of the PCA on MTU-related variables for females.

Variables	PC1	PC2	PC3	PC4
Isokinetic slow KE	–0.463	0.550	0.206	–0.315
Isokinetic fast KE	–0.156	0.392	0.728	0.080
Isokinetic slow KF	–0.209	0.478	–0.399	0.114
Isokinetic fast KF	–0.256	0.397	–0.404	0.333
CMJ height	–0.343	–0.230	0.266	0.701
DJ height	–0.423	–0.188	0.063	–0.183
DJ index	–0.416	–0.151	0.016	–0.478
RJ height	–0.435	–0.170	–0.028	0.040
RJ index	–0.425	–0.135	–0.170	–0.123
Standard deviation	2.082	1.504	0.953	0.723
Proportion of variance	0.482	0.252	0.101	0.058
Cumulative proportion	0.482	0.733	0.834	0.892
Interpretation	Strength & Jump (–)	Strength (+)/Jump (–)	KE (+)/KF (–)	CMJ & KF (+)/SSC & KE (–)

DJ, drop jump; KE, knee extension torque; KF, knee flexion torque; PC, principal component; RJ, rebound jump; SSC, stretch-shortening cycle.

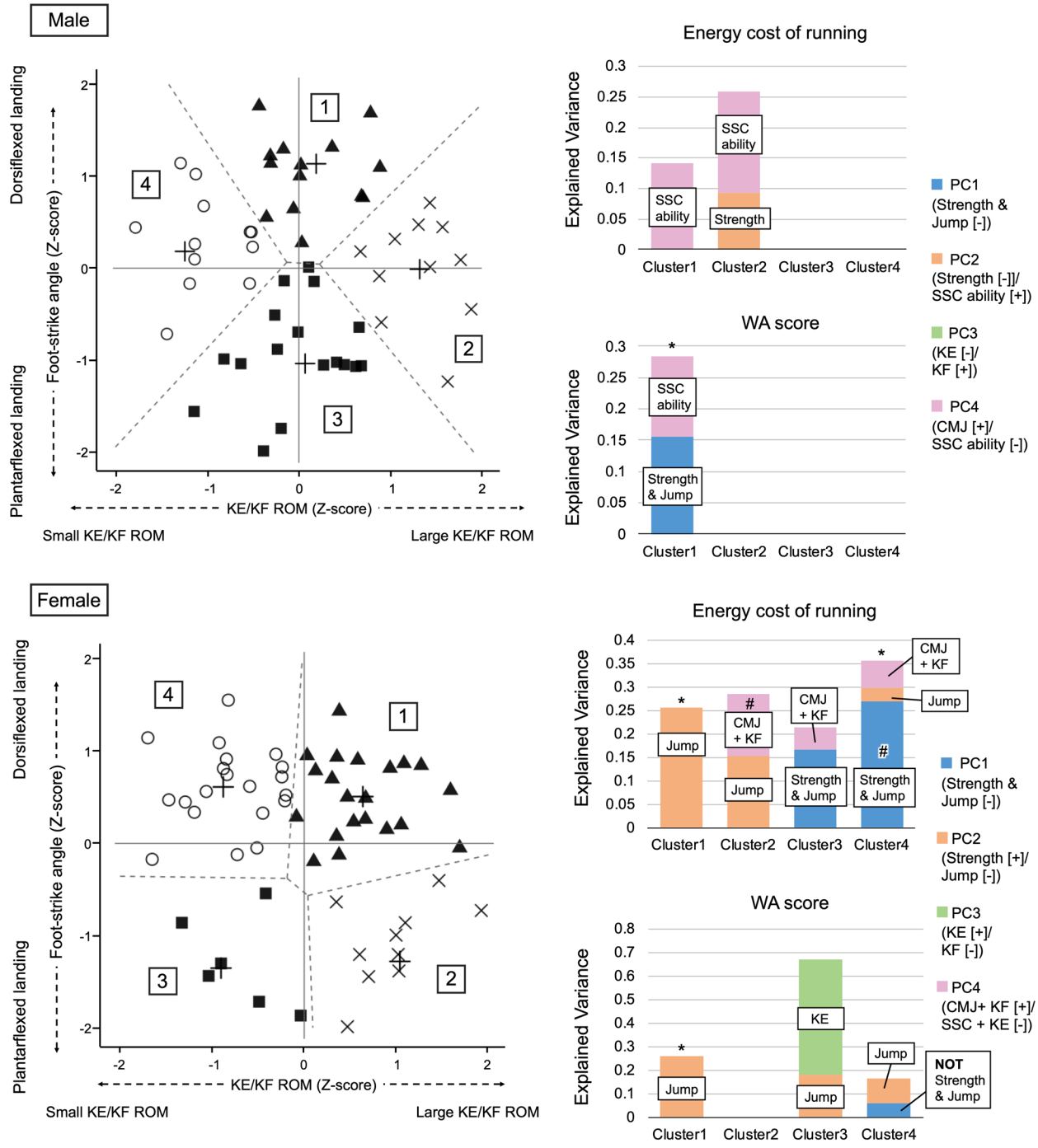


FIGURE 2—Factors associated with ECR and WA score when runners are classified based on running kinematics at 11 km·h^{-1}. Foot strike angle and KE/KF ROM during the stance phase were converted into *Z*-scores (left). The bar plots on the right depict the relative contributions of regression-selected variables using the Lindeman–Merenda–Gold method (right). The MTU function shown within each box (e.g., SSC ability) indicates that a higher value of that factor is associated with a more favorable outcome (i.e., lower ECR or higher WA score). * indicates a significant model-adjusted R^2, and # indicates a regression slope significantly different from the nonclustered analysis. "NOT" in the interpretation of PC loading indicates that a lower value is associated with a lower ECR or a higher WA score. CMJ, countermovement jump; KE, knee extension torque; KF, knee flexion torque; PC, principal component; SSC, stretch-shortening cycle.

Female Clusters at 11 km·h^{-1}

Participants were divided into four clusters in the 11 km·h^{-1} condition (Fig. 2). In the nonclustered analyses, only weak to fair relationships were observed between MTU function variables and running performance-related variables (ECR, r = −0.109 to 0.267; WA, r = −0.276 to 0.175; see "ALL" columns in Supplemental Table 3, Supplemental Digital Content, https://links.lww.com/MSS/D402).

Cluster 1: Dorsiflexed landing and large KE/KF ROM. PC2 (higher jump performance) significantly explained the variance in both ECR (adjusted $R^2 = 0.256$, $P = 0.013$) and WA score (adjusted $R^2 = 0.259$, $P = 0.013$).

Cluster 2: Plantarflexed landing and large KE/KF ROM. PC2 (higher jump performance) and PC4 (higher CMJ + KF torques) were selected as explanatory variables for lower ECR; however, these associations did not reach statistical significance (adjusted $R^2 = 0.285$, $P = 0.128$). In addition, the linear regression slope of PC4 was significantly different from that of the nonclustered runners ($P = 0.008$).

Cluster 3: Plantarflexed landing and small KE/KF ROM. PC1 (higher overall MTU function) and PC4 (higher CMJ height and KF torque) were selected as explanatory variables for lower ECR, although the model wasnot statistically significant (adjusted $R^2 = 0.214$, $P = 0.324$). Similarly, PC2 (higher jump performance) and PC3 (higher KE torque) were associated with WA score, but this relationship did not reach statistical significance (adjusted $R^2 = 0.667$, $P = 0.089$).

Cluster 4: Dorsiflexed landing and small KE/KF ROM. PC1 (higher CMJ height and KF torque), PC2 (higher jump performance), and PC4 (higher CMJ + KF torque) were significantly associated with lower ECR (adjusted $R^2 = 0.356$, $P = 0.015$). PC1 (lower overall MTU function) and PC2 (higher jump performance) contributed to the explanation of variance in WA score, indicating that jump performance rather than muscle strength was associated with WA score, while this association did not reach statistical significance (adjusted $R^2 = 0.164$, $P = 0.078$).

Male Clusters at 80% of $v\dot{V}O_{2max}$

Participants were divided into four clusters in the 80% $v\dot{V}O_{2max}$ condition (Fig. 3). In the nonclustered analyses, only weak to fair relationships were observed between MTU function variables and running performance-related variables (ECR, $r = -0.034$ to 0.115; see "ALL" columns in Supplemental Table 4, Supplemental Digital Content, https://links.lww.com/MSS/D402).

Cluster 1: Dorsiflexed landing and large KE/KF ROM. PC2 (higher SSC ability) significantly explained the variance in ECR (adjusted $R^2 = 0.483$, $P = 0.007$), and its regression slope was significantly different from that of nonclustered runners ($P = 0.016$). In addition, PC1 (lower overall MTU function) and PC2 (higher strength) were significantly associated with WA score (adjusted $R^2 = 0.553$, $P = 0.011$), indicating that muscle strength rather than jump performance was associated with WA score. The linear regression slope of PC1 significantly differed from that of nonclustered runners ($P = 0.036$).

Cluster 2: Plantarflexed landing and large KE/KF ROM. PC4 (higher CMJ height) was associated with WA score, although this relationship did not reach statistical significance (adjusted $R^2 = 0.079$, $P = 0.172$).

Cluster 3: Plantarflexed landing and small KE/KF ROM. PC4 (higher SSC ability) significantly explained the variance in both ECR (adjusted $R^2 = 0.358$, $P = 0.031$) and WA score (adjusted $R^2 = 0.383$, $P = 0.025$).

Cluster 4: Dorsiflexed landing and small KE/KF ROM. PC1 (higher overall MTU function) was selected as an explanatory variable for ECR, but it was not significant (adjusted $R^2 = 0.056$, $P = 0.161$). In addition, PC1 (higher overall MTU function) and PC4 (higher SSC ability) were associated with WA score (adjusted $R^2 = 0.098$, $P = 0.162$), although these relationships did not reach statistical significance. The linear regression slope of this model significantly differed from that of the nonclustered cohort ($P = 0.036$).

Female Clusters at 80% of $v\dot{V}O_{2max}$

Participants were divided into three clusters in the 80% of $v\dot{V}O_{2max}$ condition (Fig. 3). In the nonclustered analyses, only weak to fair relationships were observed between MTU function variables and running performance-related variables (ECR, $r = -0.137$ to 0.224; see "ALL" columns in Supplemental Table 5, Supplemental Digital Content, https://links.lww.com/MSS/D402).

Cluster 1: Dorsiflexed landing. PC1 (higher overall MTU function) and PC4 (higher CMJ height and KF torque) were significantly associated with lower ECR (adjusted $R^2 = 0.352$, $P = 0.038$). In addition, PC2 (higher jump performance), PC3 (higher KE torque), and PC4 (higher CMJ height and KF torque) significantly explained the variance in WA score (adjusted $R^2 = 0.595$, $P = 0.016$).

Cluster 2: Plantarflexed landing and large KE/KF ROM. PC2 (higher jump performance) was selected as an explanatory factor for ECR, although this relationship did not reach statistical significance (adjusted $R^2 = 0.057$, $P = 0.153$).

Cluster 3: Plantarflexed landing and small KE/KF ROM. PC1 (lower overall MTU function) and PC2 (higher jump performance) significantly explained the variance in ECR (adjusted $R^2 = 0.159$, $P = 0.047$), indicating that jump performance rather than muscle strength was associated with ECR. In addition, PC2 (higher jump performance) and PC4 (higher CMJ height and KF torque) were significantly associated with WA score (adjusted $R^2 = 0.222$, $P = 0.024$), and the linear regression slope was significantly different from that of the nonclustered cohort ($P = 0.015$).

DISCUSSION

Overall Shared Trends

In both sexes, whole-cohort analyses revealed only weak to fair correlations between MTU function and

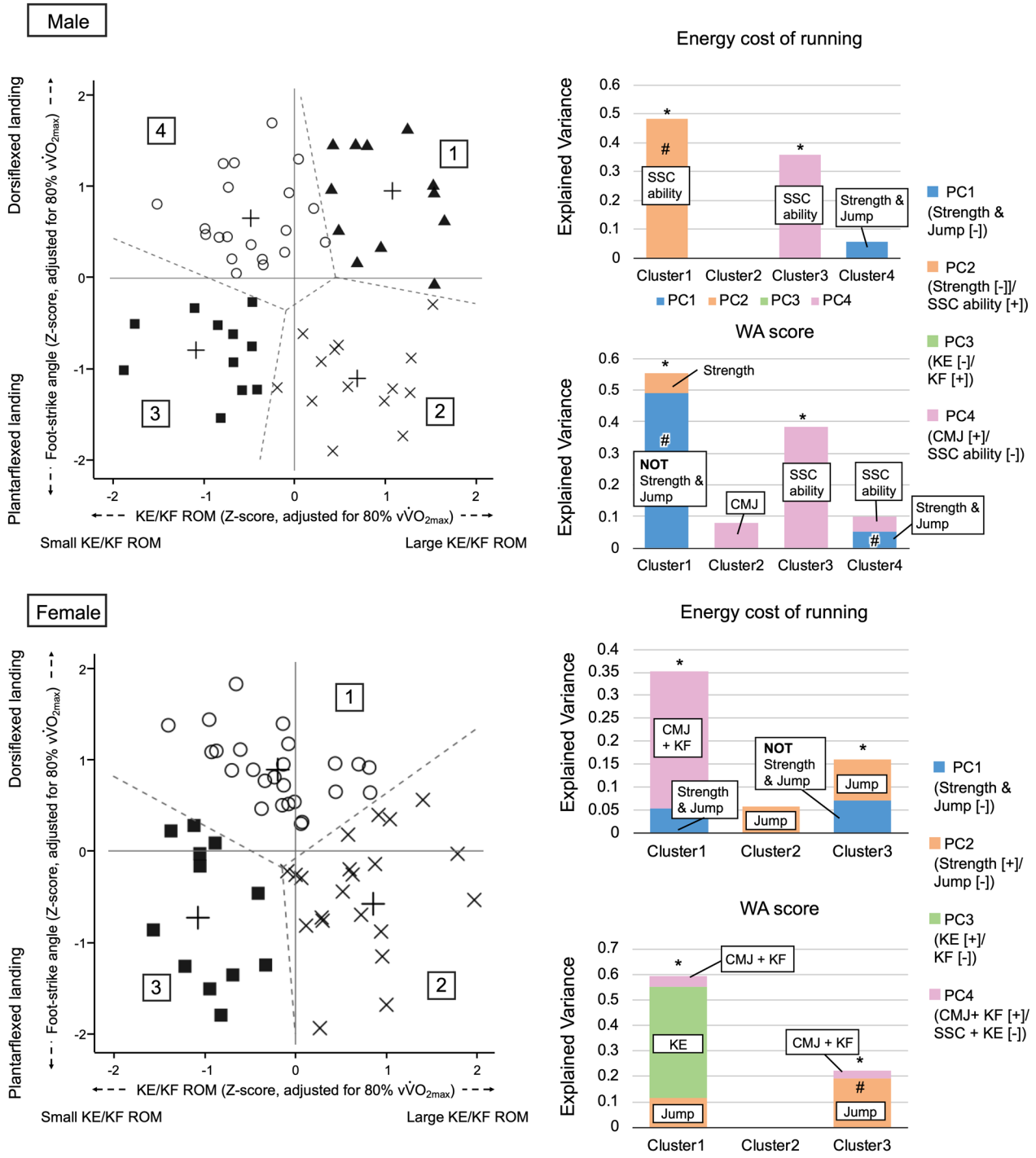


FIGURE 3—Factors associated with ECR and WA score when runners are classified based on running kinematics at 80% of $v\dot{V}O_{2max}$. Foot-strike angle and KE/KF ROM during the stance phase were converted into *Z*-scores (left). The bar plots on the right depict the relative contributions of regression-selected variables using the Lindeman–Merenda–Gold method (right). The MTU function shown within each box (e.g., SSC ability) indicates that a higher value of that factor is associated with a more favorable outcome (i.e., lower ECR or higher WA score). * indicates a significant model-adjusted R^2, and # indicates a regression slope significantly different from the nonclustered analysis. "NOT" in the interpretation of PC loading indicates that a lower value is associated with a lower ECR or a higher WA score. CMJ, countermovement jump; KE, knee extension torque; KF, knee flexion torque; PC, principal component; SSC, stretch-shortening cycle.

running economy and/or performance at both 11 $km \cdot h^{-1}$ and 80% of $v\dot{V}O_{2max}$. Several of these trends were significantly strengthened when runners were stratified into kinematics-based clusters, with the patterns varying by cluster. This finding supports our first hypothesis that clustering based on foot-strike angle in

the sagittal plane and KE/KF ROM can enhance the detection of associations between MTU function and performance-related outcomes. However, most clusters did not consistently align with the relationships hypothesized in the Background (i.e., dorsiflexed landing with KF function; large KE/KF ROM with KE function; and plantarflexed landing and small KE/KF ROM with jump performance), and these associations were not uniformly reinforced. Specifically, in male runners, significant explanatory variables accounted for 1 out of 8 clusters (12.5%) at 11 km·h^{-1}, whereas 3 out of 8 clusters (37.5%) were significant at 80% of $v\dot{V}O_{2max}$. In female runners, 4 out of 8 clusters (50.0%) reached significance at 11 km·h^{-1} and 4 out of 6 clusters (66.7%) at 80% of $v\dot{V}O_{2max}$. Accordingly, we interpret the findings by first addressing the overall trends and then considering the moderating effects of sex and running velocity.

Unexpectedly, a consistent pattern was observed across sexes and running conditions, whereby jump performance played a more important role in running economy and performance in dorsiflexed clusters than in plantarflexed clusters. This may be attributable to the greater braking impulse experienced by rearfoot strikers during the early stance phase compared with forefoot strikers (63), which can negatively affect running economy (7). Indeed, rearfoot strikers (i.e., dorsiflexed runners) have been shown to exhibit greater activation of knee-joint musculature during the early stance phase to attenuate the larger braking impulse, compared with forefoot strikers (63). Accordingly, reducing the braking phase and enhancing the efficient storage and release of elastic energy through the SSC function may be particularly beneficial for rearfoot strikers. In contrast, forefoot strikers generally exhibit greater activation and co-activation of muscles surrounding the ankle joint rather than the knee joint during running (63,64). In the present study, drop and rebound jumps may partially reflect ankle-joint musculotendinous SSC ability; however, these tasks are also characterized by a substantial contribution of knee-joint-dominant SSC function (65–67). Taken together, these findings suggest that jump performance, particularly knee-dominant musculotendinous SSC ability, may represent a limiting factor for running economy and performance in dorsiflexed runners.

Sex Differences

In this section, we focus on three key aspects of the results in female runners. First, jump performance emerged as an important determinant of both running economy and performance across multiple female clusters. Second, the overall MTU function was particularly relevant in female runners characterized by small KE/KF ROM. Third, a relatively larger number of MTU-related variables were selected in the regression models for female clusters compared with males. Together, these findings suggest a more complex and heterogeneous contribution of MTU function to running economy and performance in female runners, which is discussed below.

First, although we expected that KE torque would be associated with running economy and performance in female runners, jump performance emerged as a key factor associated with both running economy and performance in most female clusters. Sex-specific neuromuscular control strategies may partly explain this finding. Previous studies have reported that females exhibit a higher frequency of muscle co-contraction during walking (68,69) and jumping tasks (70) than males, which increases the demand for neuromuscular capacity during SSC actions. In addition, adult females tend to preferentially activate the rectus femoris to stabilize the knee joint, as this biarticular muscle contributes not only to KE torque production but also to the control of hip excursion during landing and jumping (71–73). In the present study, PCA in female runners also revealed a co-variation between drop and rebound jump performance and KE torque, particularly slow KE torque, suggesting that KE function may indirectly contribute to superior running economy and performance through its association with jump-related SSC performance. Collectively, these findings indicate that female runners may rely on a neuromuscular strategy characterized by enhanced co-contraction and quadriceps-dominant control during running, thereby increasing the importance of jump performance for running economy and performance.

Second, in female runners, most clusters characterized by small KE/KF ROM showed the importance of overall MTU function for improved running economy and performance. In contrast, in male runners with small KE/KF ROM, SSC ability was selectively associated with running economy and performance, particularly in cluster 3 at 80% of $v\dot{V}O_{2max}$. This finding suggests that greater tendon stiffness in males may facilitate rapid force transmission and modulation of muscle activation during short ground contact times (24,74). In contrast, female runners exhibited a broader dependence on MTU function, encompassing both muscle strength and jump performance. As mentioned above, females tend to rely more on muscle co-contraction to enhance joint stability (68–70) and compensate for lower tendon stiffness (74). However, this strategy may limit effective utilization of the SSC by increasing muscle fascicle work and reducing the amount of elastic energy stored and returned by tendinous tissues (75,76). Consequently, greater muscle strength is required to maintain leg stiffness, while superior jump performance becomes necessary to offset reduced SSC efficiency. Together, these findings suggest that male runners with small KE/KF ROM can benefit more selectively from SSC ability through effective force transmission, whereas female runners with small KE/KF ROM require a wider range of MTU functions to maintain running economy and performance.

Third, a relatively greater number of MTU-related variables were selected for female clusters compared with males at 11 km·h^{-1}. At a given absolute running speed, female runners typically run at a higher intensity than male runners (77). Although direct evidence remains limited, this higher relative intensity may amplify the contribution of MTU function, including both muscle strength and SSC-related performance, to running economy and performance in female runners. Indeed, available data suggest that running at higher relative intensities increases reliance on muscle force production and reactive strength, irrespective of sex (5). In addition, females may depend more on active muscle force to meet mechanical demands under conditions of higher relative intensity (78), as sex differences in neuromuscular activation and fatigability have been reported (79). Consequently, the greater number of MTU-related variables selected in female clusters may reflect not only sex-specific neuromuscular strategies but also the higher relative intensity at which female runners performed the running tasks, underscoring the importance of overall MTU function for maintaining running economy and performance. Nevertheless, the number of studies directly examining sex-specific determinants of running economy and performance across different relative intensities remains limited, highlighting a substantial need for further research in this area.

Effects of Running Velocities

Our hypothesis that the relationships of MTU function with running economy and performance would be strengthened in a cluster-specific manner was grounded in prior studies conducted at the same absolute running velocity across runners (28,29,34–36), because few studies have examined running kinematics or profiles at a matched relative velocity. Nevertheless, a greater number of MTU-related variables were associated with running economy and/or performance at 80% of $v\dot{V}O_{2max}$ than at 11 km·h^{-1}, regardless of sex, with this tendency being particularly evident in male runners. Importantly, these patterns remained after statistical adjustment for running velocity when examining the relationships between joint angles and ECR. This adjustment underscores the utility of accounting for running velocity when evaluating MTU-related determinants under relative intensity conditions.

In male runners, a clear velocity-dependent shift was observed in the associations between MTU function and running economy and performance, with substantially more significant relationships emerging at 80% of $v\dot{V}O_{2max}$ than at 11 km·h^{-1}. At the lower speed, the mechanical demands required to generate vertical and propulsive impulse appear to be relatively modest (16,78), thereby possibly reducing the sensitivity of running economy and performance to interindividual differences in MTU function. As running velocity increases, however, contact time shortens, and the temporal constraints on impulse generation become more stringent (16,78), requiring more rapid force development and more effective jump performance (Fig. 3). Consequently, impulse generation becomes increasingly dependent on active musculotendinous function, a pattern likely to be particularly pronounced in male runners for whom the difference between 11 km·h^{-1} and 80% of $v\dot{V}O_{2max}$ represents a substantial increase in relative and absolute mechanical demand. This interpretation is consistent with biomechanical evidence indicating that MTU functions become critical determinants of running economy and performance at higher running intensities.

In contrast, female runners showed no marked velocity-dependent change in the overall pattern of associations, suggesting that the fundamental muscular demands required to generate the necessary impulse were already present at both running velocities. This finding may reflect a neuromuscular strategy in which impulse generation relies more consistently on MTU functions across speeds, likely due to a greater need for muscular stabilization (71–73) and lower passive tendon stiffness (74). As a result, increasing running velocity did not introduce new limiting factors in terms of overall MTU function. Nevertheless, compared with the 11 km·h^{-1} condition, the importance of CMJ performance and KF torque in relation to ECR increased at 80% of $v\dot{V}O_{2max}$. Although the increase in running velocity was relatively modest in female runners (absolute, 11.0 km·h^{-1}; relative, 12.4 ± 0.9 km·h^{-1}), previous evidence indicates that knee flexor activation increases with running speed, even with a slight running velocity change (9.4–13.3 km·h^{-1}) (80). Moreover, females have been reported to exhibit a greater involvement of the KF activation during CMJ compared with males (81). Accordingly, even without a marked increase in the overall reliance on MTU function across speeds, relatively higher-intensity running in females may selectively amplify the contribution of knee flexor function and, secondarily, CMJ-related SSC capacity to efficient force production and braking control. This highlights sex-specific differences in how velocity-related mechanical demands are accommodated.

Limitations

This study has some limitations. First, clusters were derived from the relative positions of runners within our sample of 120 Japanese endurance runners; thus, generalizability to other ages, populations, and competition levels remains uncertain. Preliminary longitudinal observations in our cohort suggest that individual cluster membership is relatively stable over time (unpublished data), supporting the use of this approach to identify performance-relevant MTU function in similar populations. Future multicenter data pooling will be important for external validation and for developing

individualized, kinematics-informed training prescriptions tailored to sex, competition level, and population background. Second, the cross-sectional design precludes causal inference. Whether targeted strength training improves running economy and performance in a cluster-specific manner remains to be determined. Randomized, cluster-stratified training interventions are needed to establish causality and to evaluate potential adverse responses in specific kinematic profiles. Third, the timing of data collection was not standardized across participants, as measurements were conducted across different phases of the training season (e.g., off-season and in-season) (82). Running economy, muscle strength, and jump performance may fluctuate over a periodized training cycle. Although significant relationships between MTU-related variables and ECR were observed, the magnitude of these associations may differ depending on seasonal training status. Therefore, future longitudinal studies with controlled measurement timing are needed.

CONCLUSIONS

Classifying endurance runners by foot-strike angle in the sagittal plane and KE/KF ROM strengthened several relationships between MTU function (strength and jump performance) and running economy/performance compared with whole-cohort analyses, and did so in a cluster-specific manner, especially at 80% of $v\dot{V}O_{2max}$. Specifically, jump performance correlated with running economy and performance in clusters with dorsiflexed landing and female runners overall, whereas overall MTU function was associated with running economy and performance in clusters characterized by small KE/KF ROM. Notably, female runners exhibited a broader dependence on MTU-related variables, suggesting a more complex and heterogeneous contribution of muscle strength and SSC-related capacity to running economy and performance compared with males. These findings support kinematics-based clustering as a practical framework for individualizing strength training prescriptions to enhance running economy and performance, while accounting for sex-specific and kinematics-dependent variations.

The experiment was supported by JST SPRING, Grant Number JPMJSP2101. The authors declare no conflicts of interest. The results of the study are presented clearly, honestly, and without fabrication, falsification, or inappropriate data manipulation. The results of the present study do not constitute endorsement by the American College of Sports Medicine. Y. E. conceived and designed the study. All authors contributed to data collection, analysis, and/or interpretation of the results. Y. E., S. M., and H. K. drafted the manuscript. All authors read and approved the final version of the manuscript. This study was approved by the Ethics Committee of Ritsumeikan University (BKC-LSMH-2023-011). Written informed consent was obtained from each participant. The datasets generated and/or analyzed during the current study are available from the corresponding author upon reasonable request.

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